

JEE MAIN REVISION MASTER: CHEMICAL KINETICS AND NUCLEAR CHEMISTRY SYLLABUS REDUCED

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 221 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Chemical Kinetics And Nuclear Chemistry - Core Application	221 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Chemical Kinetics And Nuclear Chemistry - Core Application (Very High Yield — 221 PYQs)

Core Principle: Key problem pattern in Chemical Kinetics And Nuclear Chemistry.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : $R=8.314\text{JK}^{-1}\text{mol}^{-1}$ and $1\text{cal}=4.2\text{J}$ Statement I : When $E_a=12.6\text{kcal/mol}$, the room temperature rat..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Consider the reaction $aX \rightarrow bY$, for which the rate constant at 30°C is $1 \times 10^{-3}\text{mol}^{-1}\text{Ls}^{-1}$. Which of the ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "At 27°C in presence of a catalyst, activation energy of a reaction is lowered by 10kJmol^{-1} . The logarithm of ratio of $\frac{\dots}{\dots}$ " [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 24th January Morning Shift (Q#1)

At 27°C in presence of a catalyst, activation energy of a reaction is lowered by 10kJmol^{-1} . The logarithm of ratio of $(k(\text{catalysed}) / k(\text{uncatalysed}))$ is.... (Consider that the frequency factor for both the reactions is same)

- (A) 1.741 **[Correct]**
- (B) 0.1741
- (C) 17.41
- (D) 3.482

Correct Answer: A

Solution Analysis: Arrhenius equation (NCERT): $k = Ae^{-\frac{E_a}{RT}}$ Given: catalyst lowers activation energy by $10\text{kJ mol}^{-1} = 10000\text{J mol}^{-1}$ and frequency factor A is same. So,
 $\frac{k_{\text{cat}}}{k_{\text{uncat}}} = \frac{Ae^{-\frac{E_{a,\text{cat}}}{RT}}}{Ae^{-\frac{E_{a,\text{uncat}}}{RT}}} = e^{\frac{E_{a,\text{uncat}} - E_{a,\text{cat}}}{RT}} = e^{\frac{10000}{RT}}$
 At 27°C, $T = 300\text{K}$ and $R = 8.314\text{J mol}^{-1}\text{K}^{-1}$: $\frac{10000}{RT} = \frac{10000}{8.314 \times 300} \approx 4.009$ So,
 $\log_{10}\left(\frac{k_{\text{cat}}}{k_{\text{uncat}}}\right) = \log_{10}(e^{4.009})$